

STUDENT BOOK ANSWERS

Chapter 2 Classifying and separating substances

Question set 2.1

1	Mixture	Pure substance
	<i>Heterogeneous</i>	Homogeneous
	Variable composition	<i>Constant composition</i>
	At least two different types of particles	<i>Only one type of particles</i>
	<i>Variable properties</i>	Constant properties

2 a i Copper: homogeneous

ii Soil: heterogeneous

iii Air: homogeneous

iv Salt water: homogeneous

v Carbon dioxide: homogeneous

vi Salt: heterogeneous

vii Nitrogen: homogeneous

viii Lemonade: heterogeneous

ix Blood: heterogeneous

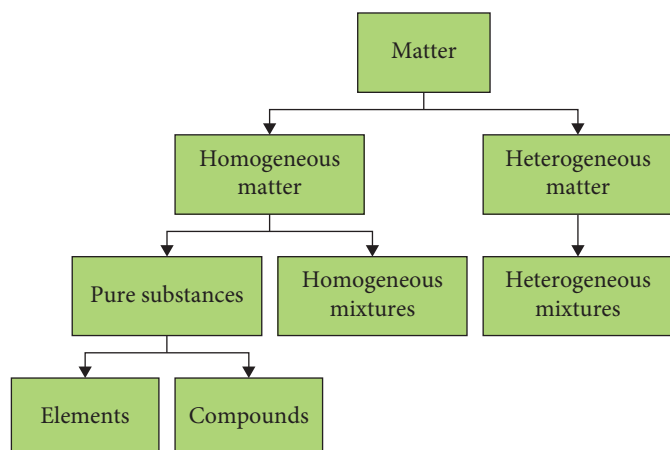
x Mercury: homogeneous

b i Element, iii mixture, iv mixture, v pure substance, vii element, x element

3 A solution is a solute dissolved in a solvent, a combination of two different substances. A mixture can be a solid, liquid or a gas or a combination of these rather than just a liquid.

4 Whipped cream is air trapped in cream. Cream itself is a high fat and protein emulsion the consistency through the mixture is homogeneous.

5 Students' answers will vary.



- 6 a C, conducts heat, but not malleable
 b B, conducts electricity
 c A, is transparent but does not conduct heat

Experiment 2.1: Observing chemical and physical properties of pure substances

Results

1	Substance	Physical state	Colour	Odour	Solubility in water	Effect of magnet
	Sulfur	<i>Solid</i>	<i>Pale yellow</i>	<i>No</i>	<i>No</i>	<i>None</i>
	Iron filings	<i>Solid</i>	<i>Greyish</i>	<i>No</i>	<i>No</i>	<i>Yes</i>
	Sodium hydrogen carbonate	<i>Solid</i>	<i>White</i>	<i>No</i>	<i>Yes</i>	<i>No</i>
	Sodium chloride	<i>Solid</i>	<i>White</i>	<i>No</i>	<i>Yes</i>	<i>No</i>
	Sucrose	<i>Solid</i>	<i>White</i>	<i>No</i>	<i>Yes</i>	<i>No</i>
	Magnesium	<i>Solid</i>	<i>Greyish</i>	<i>No</i>	<i>No</i>	<i>No</i>
	Copper chloride	<i>Solid</i>	<i>Brown solid, or a blue-green.</i>	<i>No</i>	<i>Yes</i>	<i>No</i>
	Aluminium	<i>Solid</i>	<i>Greyish</i>	<i>No</i>	<i>No</i>	<i>No</i>

2	Substances	Observations
	Fe and S – tested with magnet	<i>Fe would be attracted by the magnet and separate from the sulfur</i>
	Mg and HCl	<i>Bubbles formed and Mg strip observed to disappear. Heat given off</i>
	Sucrose and HCl	<i>No reaction</i>
	Sodium hydrogen carbonate and HCl	<i>Bubbles observed and solid appears to disappear</i>
	Sodium chloride and HCl	<i>Solid appears to dissolve</i>
	Copper(II) chloride and HCl	<i>Greenish solution formed</i>
	Al and HCl	<i>Bubbles appear and the metal appears to disappear</i>
	Sucrose – heated	<i>Material melts, starts to become light brown and odour of toffee is apparent. Upon further heating it turns black and produces wisps of smoke.</i>
	Al – heated	<i>Substance appears to melt and become liquid</i>
	Copper(II) chloride and Al	<i>Heat given off. Al appears to disappear and brownish material is seen</i>

Analysis of results

1 No change

Chemical change

No change

Chemical change

Physical (solid – aqueous)

Physical (solid – aqueous)

Chemical change

Chemical change

Physical change

Chemical change

2 Bubbles observed and heat given off. Magnesium appears to disappear.

Bubbles observed, solid disappears.

Bubbles observed and heat given off. Aluminium appears to disappear.

Sucrose melts and changes to liquid but then with vigorous heating will form soot and wisps of smoke.

Heat given off, aluminium appears to disappear and reddish brown substance appears.

3 a Heated aluminium looks the same colour as aluminium but is now liquid.

b Sucrose is a solid white material. Heated sucrose will turn light brown and liquidify.

4 Aluminium is chemically very reactive with acids, sucrose is less chemically reactive. Upon heating it decomposes.

Conclusion

Students' answers will vary.

Question set 2.2

1	Physical change	Chemical change
	<i>No new substances formed</i>	New substance formed
	Easy to reverse	<i>Hard to reverse</i>
	<i>Small energy change</i>	Generally large energy change
	Generally physical properties don't change	<i>Physical properties may change</i>
	Mass conserved	<i>Mass conserved</i>

- 2 a Melting point: physical
b Ductility: physical
c Reactivity: chemical
d Solubility: physical
e Flammability: chemical
f Colour: physical
g Odour: physical
h Catches fire in air: chemical
- 3 Colour change, gas produced, solid (called a precipitate) is produced or 'disappears' or a significant change in temperature, odour produced
- 4 a Fading of dye in cloth: chemical
b Melting of ice: physical
c Digestion of food: chemical
d Dissolving sugar in tea: physical
e Burning a candle: chemical
f Making a cake: chemical
g Evaporating water: physical
h Producing light in a glow stick: chemical
- 5 a roasting the seeds → crushing the seeds → washing → baking
b chemical, physical, physical, chemical
c Toxins broken down, breaking into smaller size, removing soluble compounds by dissolving in water, chemical changing of the seeds
- 6 It is not correct to say that helium has no chemical properties. Helium has low reactivity, and a stability which makes it less likely to react with other substances.
- 7 Students answers will vary. They may consider: the presence or absence of salt, measure the time taken to boil. Control variables such as amount of water, heat source and container.

Question set 2.3

- 1 The larger the holes in a sieve, the quicker the material will pass through. If the holes are too large then all the material may pass through making the sieve ineffective as no separation occurs. If too small then no material will pass through which is also ineffective. The size of the holes has to be between the sizes of the materials being separated.
- 2 Distillation relies on differences in boiling point. Simple distillation is when the lower boiling point substance is boiled off from the main bulk of the materials. Fractional distillation is a more sophisticated distillation in which there is more accurate separation of the different components, resulting in many 'fractions' or different parts of the material being collected.

	Separation method	Property enabling separation
Separating mud from water	<i>Filtration</i>	<i>Difference in state and size of particles</i>
Separating salt and sand	<i>Sieving</i>	<i>Difference in particle size</i>
Separating oil and water	<i>Separating funnel</i>	<i>Difference in densities</i>
Obtaining nitrogen and oxygen from liquid air	<i>Fractional distillation</i>	<i>Difference in boiling points</i>
Obtaining fresh water from salt water	<i>Distillation</i>	<i>Big differences in boiling point</i>
Separating gold from soil	<i>Sieving (then probably panning)</i>	<i>Sieving: Differences in particle size Panning: Differences in density</i>

- 4
 - a Students' answer will vary. Place in a separating funnel to remove the sand from the other material → magnetic field (use a magnet) to remove the Ilmenite → place in an electrical field to separate the rutile from the zircon
 - b Students' answer will vary.
- 5
 - a 1 = Sieving; sieve; 2 = Separation by density; separating funnel; 3 = Distillation/evaporation; distillation apparatus
 - b X = iron; Y = sugar; Z = water
 - c Use the fractional distillation technique.

Chapter review questions

- 1 Mixture: can be separated into two or more pure substances by physical or mechanical means, has properties that may change as the relative amounts of the substances present are changed, different parts of the mixture show different properties, has a variable composition, may be homogenous or heterogeneous.

Pure substance: cannot be separated by physical or mechanical means, is homogeneous, properties are constant and do not change, has a fixed composition.
- 2 Physical change is when no new substance is formed, easily reversed, relatively small energy change.
Chemical change is when at least one new substance is formed, difficult to reverse, and generally a large energy change.
- 3
 - a Density: physical
 - b Boiling point: physical

- c Reaction with oxygen: chemical
 - d Solubility: physical
- 4**
- a Filtration: Difference in state and size of particles
 - b Fractional distillation: Significant but small difference in boiling points
 - c Electrostatic separation: Difference in electrical charge
 - d Vaporisation: Solvent has a much lower boiling point than the dissolved solid
- 5**
- a Paint: solution
 - b Tap water: solution
 - c Oxygen: element
 - d Honey: solution
 - e Granite: heterogeneous mixture
 - f Lead sheeting: element
- 6**
- a Burning a piece of coal: chemical
 - b Tearing a piece of paper: physical
 - c Tarnishing of silver: chemical
 - d Fruit rotting: chemical
 - e Dissolving instant coffee in hot water: physical
- 7** Lack of reactivity should be considered as a chemical property as it indicates how it behaves chemically with other substances.
- 8** A change in which no new chemical is formed is called a physical change; The substance has had a change in its appearance, or change of state, or some other physical property has changed such as electrical conductivity.
- 9** Solubility; magnetic attraction; evaporation point; state of matter at room temperature
- 10**
- a Salt and iron filings: magnetism – iron filings are magnetic; salt is not; or solubility – salt should dissolve in water and iron will not
 - b Iron filings and aluminium filings: magnetism – iron is magnetic and aluminium is not
 - c Sugar and water: evaporation – water has a lower boiling point than sugar
 - d Oil and vinegar: density – oil and vinegar are insoluble and could be separated by a separating funnel
- 11**
- a A chemical reaction has occurred, so the liquids have changed into new substances.
 - b The precipitate could be separated by filtering the sample.
- 12** Both glass and some plastics are transparent so can be used in applications where visibility is important, and both are unreactive making them suitable for containers.
- 13** Once oxygen has reacted to form a compound the properties of the material are different. The element is a gas but when it reacts with elements like iron it forms a solid.